

MATH 2790 Fall 2025

Homework 8

Due date: Wednesday, Oct 22

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 3.8–3.9.

Problems:

1. Question (1) on page 188 of textbook under section 3.8.
2. Mike purchases a perpetuity providing a series of increasing payments. The first payment of 1200 is paid in a year and each payment is 2% larger than the preceding payment. Find the present value of the perpetuity given the annual effective interest rate of 6%.
3. Chris makes annual deposits into a bank account at the beginning of each year for 20 years. Chris's initial deposit is equal to 100, with each subsequent deposit $k\%$ greater than the previous year's deposit. The bank credits interest at an annual effective rate of 5%. At the end of 20 years, the accumulated value in Chris' account is equal to 7276.35. Given $k > 5$, set up the equation of k . Use a graphing device to estimate the value of k .
4. Find the present value of an annuity-due with 17 annual payments of 100, 96, 92, $\dots$, 36 with the annual effective interest rate of 5%.
5. Olga buys a 5-year increasing annuity for X . Olga will receive 2 at the end of the first month, 4 at the end of the second month, and for each month thereafter the payment increases by 2. The nominal interest rate is 6% convertible monthly. Calculate X .
6. Question (3) on page 189 of textbook under section 3.9.
7. Olga buys a 5-year increasing annuity for X . Olga will receive 2 at the end of the first month, 4 at the end of the second month, and for each month thereafter the payment increases by 2. The nominal interest rate is 6% convertible monthly for the first two years and 4.8% convertible monthly for the last three years. Calculate X .